

Capturing Images with USB Camera, Wi-Fi and Raspberry Pi

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Ever since this tiny, credit-card-sized computer, Raspberry Pi, appeared in the market, it has caught the imagination of every electronics and computer hobbyist around the world. The powerful Linux operating system combined with 26 input-output (I/O) pins can do many amazing things out-of-the-box. This article explains how to interface a USB camera and Wi-Fi with Raspberry Pi. It also explains the steps for making a webcam server using VLC Apache2 server software.

Raspberry Pi operating system

Several formats of Linux are available for installation on this tiny board. Among these, the most popular ones are Archlinux, Wheezy Linux, qonpi, Fedora and xbmc.

I prefer Wheezy Linux because it is Debian style, where installation and tweaking is quite easy. The 2014-09-09-wheezy-raspbian.zip is available for download on raspberrypi.org. The wireless adapter, which comes in the form of a Wi-Fi dongle, can be installed out-of-the-box, so can be the USB cameras.

The steps to interface a USB webcam and Wi-Fi with a Raspberry Pi (model-B) board are given below. The USB webcam interfaced to a Raspberry Pi board is shown in Fig. 1.

Preparing the operating system. After downloading the zip file (wheezy-raspbian), extract the image in a folder on your computer and insert a 2GB or bigger-size SD card in the card reader slot. At this stage, check the identity of the SD card—sda, sdb or sdc—as shown in your

Linux PC. After inserting the SD card, issue an 'ls /dev/sd*' command at a terminal prompt and see what name has been given to the card.

```
bera@Acer-AOD255E:~$ ls /dev/sd*
/dev/sda /dev/sda2 /dev/sda4 /dev/sda6
/dev/sda1 /dev/sda3 /dev/sda5 /dev/sda7
/dev/sdb1 /dev/sdb2 /dev/sdc
```

Here, my SD card has been identified as /dev/sdc. Now, go to the folder where you have extracted 2012-09-18-wheezy-raspbian.img file. This image file now needs to be written on the SD card to make it work as the operating system. Any mistake here will lead to the loss of



Fig. 1: USB camera with Raspberry Pi

entire data on your computer hard disk.

Look for the highest order, like sdc, sdd, etc., to identify your SD card. To be absolutely sure, just unplug and re-plug the card a few times and notice the changes.

At the terminal prompt, issue the following command:

```
bera@Acer-AOD255E:~/Desktop/Raspberrypi/OS$ sudo dd if=2012-09-18-wheezy-raspbian.img of=/dev/sdc bs=2M
```

This will use dd, the powerful image-creator program of Linux, and transfer the image to the SD card.



When it prompts for 'su' password, enter it. (In Windows, there are several graphical image-creator programs like Nero Burner that can be used in a similar fashion.)

The above command may take several minutes. Look for HDD Lamp Busy indication. When the lamp stops blinking, go to the desktop and safely remove the SD card. (Right-click on the SD card and then click on Safely Remove.)

Connecting the display/monitor.

Connect the TV screen or monitor to Raspberry Pi board using a composite video cable (AV cable) or HDMI cable, insert the SD card into the SD card slot and switch-on the power supply.

The TV screen/monitor even on the right AV channel may refuse to come up the first time. When that happens, remove the power-supply cord of Raspberry Pi

board once (keeping the monitor's power supply on) and then re-plug it. Now, the monitor should come on live, if the SD card is not removed.

Once the Raspberry Pi starts booting up, it will prompt you for login name and password. The login for Wheezy linux is pi and the password is raspberrypi. Once entered, Raspberry Pi will open the text screen. Enter Startx command at the prompt to start the beautiful GUI desktop screen (Fig. 2). The first time when I saw the image on